SMC-BASED BAYESIAN VARIABLE SELECTION: FIVE RESEARCH CANDIDATES

Prepared 2026-10-05 for the Local_to_global_Bayesian project

Status: literature-grounded proposals, not newly benchmarked algorithms. No simulations, numerical experiments, or statistical validation were executed in this research session. Existing saved results were inspected. Future computation belongs on Owls under the user's instructions. The bibliography, source versions, reading records, and SHA-256 manifest accompany this document.

RECOMMENDATION

Keep posterior approximation and the selection decision as separately auditable parts of one SMC-based procedure. The five candidates below change either the decision or a completely specified prior; none replaces the final posterior with the composite likelihood, a variational distribution, or a screened posterior. Candidates 1, 3, 4, and 5 can use the same fitted SMC posterior. Candidate 2 changes the posterior target and therefore needs its own exact low-dimensional reference.

My initial implementation order would be 1 and 3 on existing posterior outputs, then 2 as a controlled prior experiment. Candidate 5 is a strong companion when group localization is a scientifically acceptable answer. Candidate 4 is a lower priority predictive branch because predictive projection has already been tried locally and its objective is less aligned with individual support recovery.

The literature supports the ingredients and objectives. It does not establish that these particular SMC adaptations beat the current strongest BVS benchmarks under this project's data-generating mechanisms. That is the empirical question, not a conclusion of this review.

WHAT THE EXISTING METHOD AND RESULTS ACTUALLY ESTABLISH

The production model is Gaussian regression without an intercept, with centered, sample-standardized X and uncentered y. The control covariance of beta_S given sigma^2 is sigma^2 n (X_S'X_S + I)^(-1), the variance prior is proper inverse Gamma(.01,.01), and indicators have independent Bernoulli(.2) priors. Marginal likelihoods of specified subsets are evaluated analytically using Cholesky.

Blocks of five predictors give a product of local marginal likelihoods and an exactly sampleable initial distribution with known normalizer. The bridge moves from that distribution to the full-model posterior, not to a composite-posterior approximation. The audited production configuration uses 10,000 particles, adaptive temperature selection at incremental ESS .90, systematic resampling, and one systematic coordinate Gibbs sweep per stage. Final conditional inclusion averages produce Rao-Blackwellized PIPs. Joint events require joint particles or a valid joint Rao-Blackwellization; multiplying the RB PIPs is not a substitute.

In Experiment_2_BVS_Rerun, the strong-block S4 scenario had 29/100 exact support recoveries under both exact-PIP and RB-PIP median selection. BFDR at 10% recovered 21/100 under both, and BFDR at 5% recovered 12/100 under both. Thus these results point primarily to a posterior/decision limitation, rather than an inaccurate marginal ranking caused by SMC. This is an inference from the inspected results, not an impossibility bound on any selector.

Experiment_1 reports median absolute relative Z errors of about .66%-1.73% across scenarios, with a worst observed error of 6.992%. Median RB-PIP MAE ranges from .00005229 to .0007992. Those are strong results, but not every preregistered criterion passed: S3 failed the mean-Z-ratio confidence-interval criterion and S4 failed the plain-PIP median-model agreement criterion. All bookkeeping checks passing is not the same as all statistical acceptance criteria passing.

The historical manuscript already explores F1 decisions, block MAP, independent slabs, Beta priors, pMOM, predictive projection, and resolution-based reporting. Those broad ideas should not be presented as newly discovered here. The specific extensions below are the joint loss, normalized mixture construction, particular decision paths, and explicit computational restrictions. Historical small-run improvements are exploratory; historical pMOM/independent-slab evidence failures make normalization and target-specific accuracy essential for candidate 2.

For a fixed posterior, the median rule minimizes posterior expected symmetric Hamming loss. A top-PIP prefix is optimal for maximizing expected true positives under an ordinary, equal-cost variable-level BFDR constraint. These elementary decision facts mean that using dependence cannot uniformly improve the same posterior objective. A useful improvement must change the loss, the posterior, the reporting resolution, or the tradeoff being evaluated.

1. SMC WITH JOINT LOCAL-CONFIGURATION DECISIONS

Source: Chandra and Bhattacharya (2019), Non-marginal decisions: A novel Bayesian multiple testing procedure, Electronic Journal of Statistics 13(1), 489-535, DOI 10.1214/19-EJS1535. The complete 36-page arXiv v5 author version, including supplement, was read; the final journal PDF was not retrieved. https://arxiv.org/pdf/1310.5966v5

Idea. Replace independent thresholding by a decision loss that rewards a selected variable when the decisions concerning its designated neighbors are also right. This uses posterior configuration probabilities rather than only marginals. It can select a coherent local pattern even when its individual variables have inclusion probabilities below .5. It can also reject a pattern of high marginal probabilities that has weak joint support. Neither change is uniformly beneficial.

Concrete adaptation. Partition predictors into disjoint decision groups B of bounded size b, specified from design structure or a prespecified X-based rule. They need not equal the blocks used for the SMC initial distribution. For each local pattern a subset of B, estimate

```
p_B(a) = P(S intersect B = a | y)
       by sum_i W_i 1{S_i intersect B = a}.
```

For a prespecified loss coefficient c in (0,1), choose

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a_B* = argmax over a subset of B: |a| [p_B(a) - c],
```

with the empty pattern assigned score zero, and return the union of a_B*. This follows from the source's objective sum_j d_j [w_j(d)-c]: when G_j=B, every selected variable in B has w_j(d)=p_B(a). The maximization separates across disjoint DECISION groups even if the posterior has dependence across groups. No factorization of the posterior is being assumed.

This is not block MAP: block MAP maximizes p_B(a), whereas this action trades joint correctness against the number of discoveries. The bounded disjoint case is related directly to the source's block formulation; it is not a claim to invent a new loss. The SMC implementation and its practical restrictions are the proposed synthesis.

Cost. Dense pattern counting takes O(Np), followed by O((p/b) 2^b) work per c. The exponential factor is in a deliberately bounded b, never in p. Storing all pattern probabilities costs O((p/b) 2^b); sparse maps can be smaller. This avoids the source's general overlapping-group stochastic optimization, whose original implementation used a long simulated-annealing run.

Limitations. Large groups can make every pattern unlikely and suppress power. Poor grouping misses or misrepresents important relationships. The coefficient c is a loss parameter, not a nominal FDR level. Evaluate ordinary BFDR using PIPs separately. The modified error 1-p_B(a) is at least 1-PIP_j for each selected j, but matching this modified criterion to a desired frequentist FDR requires additional analysis. The source's consistency results do not automatically transfer to this regression design or to finite-particle estimates.

Why include it. It adds a computationally explicit individual-variable selector that uses information discarded by marginal thresholding, while leaving Z, the target, and RB PIPs unchanged. Exact low-p configuration probabilities can cleanly separate a poor decision loss from Monte Carlo error.

2. SMC WITH A PROPER MIXTURE OF CONJUGATE SLABS AND A CALIBRATED SIZE PRIOR

Sources: Krishna, Bondell and Ghosh (2009), Bayesian variable selection using an adaptive powered correlation prior, Journal of Statistical Planning and Inference 139(8), 2665-2674, DOI 10.1016/j.jspi.2008.12.004. Complete author manuscript read through PMC, including all tables, appendix and references. https://pmc.ncbi.nlm.nih.gov/articles/PMC2772159/

Scott and Berger (2010), Bayes and empirical-Bayes multiplicity adjustment in the variable-selection problem, Annals of Statistics 38(5), 2587-2619, DOI 10.1214/10-AOS792. Complete 34-page electronic reprint read. https://www2.stat.duke.edu/~berger/papers/EB-multiplicity.pdf

Idea. A highly accurate approximation cannot fix a posterior that allocates support poorly for the scientific objective. Integrate over a small, proper set of plausible Gaussian slab geometries rather than choose one by maximizing the observed evidence. Study the model-size prior as a SEPARATE ablation, because its effect is different from changing coefficient geometry.

Concrete slab. Let R_S=(X_S'X_S+I)/n. Component r has covariance

```
beta_S | sigma^2,r,S ~ N(0, sigma^2 V_Sr),
V_Sr = c_r R_S^(xi_r),  c_r>0,
P(r)=omega_r, sum_r omega_r=1.
```

Start with a small fixed mixture containing the exact control component (xi=-1,c=1) and an independent Gaussian component (xi=0, chosen c). A broader predeclared geometry/scale grid is possible. The ridge ensures positive definiteness even for rank-deficient subsets. Positive powers tend to favor similar coefficients along strongly correlated directions; they must not be assumed helpful for opposite-sign cancellation. The source's power parameter is a COVARIANCE exponent; using a precision exponent reverses its interpretation.

Every component has a closed-form marginal likelihood m_r(S). With A_r=X_S'X_S+V_Sr^(-1), z=X_S'y, and a_sigma,b_sigma the proper IG parameters,

```
m_r(S) = (2*pi)^(-n/2) |V_Sr|^(-1/2) |A_r|^(-1/2)
         * b_sigma^a_sigma * Gamma(a_sigma+n/2) / Gamma(a_sigma)
         * [b_sigma + (y'y-z'A_r^(-1)z)/2]^(-a_sigma-n/2).
```

Handle S empty with the usual empty determinant and quadratic-form conventions. The target is q_1(S)=p_1(S) sum_r omega_r m_r(S). Summing a fixed finite mixture is exact for that specified discrete prior, not exact integration of an unspecified continuous hyperprior. Retain every prior normalizing constant.

Concrete size-prior ablation. Let h~Beta(a,b_p), indicators independent given h. Then

```
p_1(S) = B(a+|S|, b_p+p-|S|) / B(a,b_p).
```

This is the probability of ONE specified model, not the probability of its size; the size distribution has an additional binomial coefficient. If a prior mean size m0 is scientifically defensible, b_p=a(p/m0-1) gives E|S|=m0 for 0<m0<p. Do not set m0 to the known simulated truth. For illustration only, p=30,m0=6,a=1 gives b_p=4, matching the control's mean size at that dimension while changing the distribution. A proper, prespecified mixture over plausible m0 values is another option. Beta(1,1) by itself has mean p/2 and is not a sparse high-p rule. For adding a variable to size k, the prior ratio is (a+k)/(b_p+p-k-1).

Exact SMC start. A nonfactorized size prior does not require abandoning a known initial normalizer. Choose a normalized, factorized reference prior p_ref and positive local component scores m_br. Define

```
q_0(S) = p_ref(S) sum_r omega_r product_b m_br(S intersect B_b),
Z_br = sum over a subset of B_b p_ref,b(a) m_br(a),
Z_0 = sum_r omega_r product_b Z_br.
```

Sample r proportional to omega_r product_b Z_br, then sample blocks independently conditional on r. This produces the EXACT marginal q_0/Z_0. Bridge using q_phi=q_0^(1-phi) q_1^phi, evaluating the full mixture densities. This incorporates the changed prior ratio automatically. It is incorrect simply to substitute a dependent model prior into the old product-Z0 formula. The initial distribution need not have the same prior as q_1, but its support must cover the target.

Cost and risks. With K fixed components and subset size k, the basic cached score cost is of order K k^3 when the relevant Gram matrices are available; computing new active Gram matrices also costs work depending on n. There is no enumeration over 2^p models in production. A new target can nevertheless make the bridge and mutation substantially harder. The 2009 paper used small-p searches and an additional determinant-weighted model prior; neither its exhaustive search nor an intractable prior normalizer should be imported into this design. The ridge and finite proper mixture here are adaptations, not that paper's exact algorithm.

Study a 2x2 ablation: fixed versus mixed slab, fixed versus calibrated hierarchical size prior. Apply the same prespecified decision rule to each first. Mixture and size-prior changes need not improve weak-signal recovery; they can reduce it. No benefit is credible unless Z and PIPs pass against the corresponding NEW posterior. Historical capped pMOM failures are a reason to favor this conjugate route initially, not proof that all nonlocal priors or SMC schemes must fail.

3. SMC WITH BAYESIAN DECOUPLING, ESPECIALLY BD-FD

Source: Li, Tokdar and Xu (2025), A Bayesian decision-theoretic approach to sparse estimation, arXiv:2502.00126v1. Complete 22-page paper read. This is a preprint; the inspected author publication page still lists it as such. https://arxiv.org/pdf/2502.00126v1

Idea. Approximate the full posterior first, then choose a sparse coefficient action that preserves its fitted signal. Use posterior uncertainty to weight the selection penalty. This can change WHICH correlated predictors are selected, not just how many fall above a PIP threshold.

Concrete adaptation. Compute the full posterior coefficient mean beta_bar from SMC models and analytic conditional coefficient means, embedding each into p coordinates. Use RB PIPs pi_j in the paper's false-discovery-weighted action:

```
b_lambda = argmin_b ||X beta_bar - X b||^2
                     + lambda sum_j (1-pi_j)|b_j|.
```

Return supp(b_lambda), using a prespecified numerical support tolerance. This is a weighted lasso on the SMC posterior fitted mean. No posterior is refitted or replaced by the sparse action. The same Z and PIPs remain the primary outputs. An inverse-signal penalty is another source variant, but an epsilon>0 is essential with an exact spike at zero; it is not necessary for the initial BD-FD candidate.

Choose lambda by the paper's posterior benchmarking convention or a prespecified out-of-sample rule. The benchmarking rule compares posterior intervals for the sparse action's squared predictive discrepancy with the reference action's posterior expected discrepancy. Its central 90% interval is a tuning convention, not a 90% guarantee about support recovery. Tune no parameter on known test truth.

Cost. A finite weighted-lasso path avoids subset enumeration. Coordinate passes can use X directly, with typical O(np) work per full pass; iterations, path length, and dense versus sparse data determine actual cost. There is no requirement to form a full p-by-p Gram matrix. Posterior predictive uncertainty can be computed using conditional draws from the SMC mixture where required.

Limitations. FD in the penalty name does not supply FDR control. Convexity alone does not guarantee unique coefficients with collinearity or p>n. The source uses different reference priors, and its difficult p=2000,rho=.95 example reports BD-FD TPP .40 with FDP .72. That is a material failure case, not a method to advertise as uniformly superior. Its correlated simulations also use equal-sign signals and do not reproduce this project's cancelling-pair design.

Why include it. It is a cheap, clearly specified individual support action under the existing SMC posterior. The new test is the uncertainty-weighted BD action; generic projection has already been tried locally and is not itself a novelty.

4. SMC WITH PREDICTIVE ACCEPTABLE-SUBSET SELECTION

Source: Kowal (2022), Bayesian subset selection and variable importance for interpretable prediction and classification, JMLR 23(108), 1-38. Entire paper, appendices and references read. https://www.jmlr.org/papers/v23/21-0403.html

Idea. Report the smallest subset whose predictive performance is acceptable relative to a strong posterior predictive reference, while retaining a family of competitive subsets. This targets interpretable prediction rather than claiming the single most probable support is the uniquely correct explanation.

Concrete adaptation. The SMC posterior supplies the reference predictive mean and uncertainty. For candidate subset a and prediction weights Omega, project the reference mean y_bar using

```
b_a = (X_a' Omega X_a)^(-1) X_a' Omega y_bar,
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or a prespecified generalized-inverse/regularized action if rank deficient. This is not the coefficient posterior mean conditional on S=a. Compare posterior predictive out-of-sample losses, then choose the smallest acceptable candidate.

Bound the decision family using unique terminal SMC models, a budgeted set of local neighbors, and supports proposed by candidate 3's path. Restricting this DECISION family must not remove models from the SMC target. It also means that the best-subset claim is only relative to that candidate family. The original paper uses branch-and-bound after screening to 35 variables in its large-p application; that is not a general high-p global optimization guarantee.

Predictive evaluation can use fold refits with SMC or diagnosed fold-deletion importance weights. If such weights degenerate, refit rather than silently trust them. The prior here depends on X and n: define whether folds preserve the same prior and covariate design before using deletion identities. Changing that prior requires its ratio in the importance calculation or a separately specified fit.

The source's default eta=0 and epsilon=.10 asks for probability at least .10 of beating the benchmark loss. It does NOT mean 90% confidence of no loss. Specify the probability and predictive tolerance in scientific terms before comparing methods. Acceptable-family inclusion frequencies are not posterior inclusion probabilities, and the acceptable family is not a credible set for true support.

Cost and limitations. With a bounded family, predictive fitting and scoring have polynomial costs in the candidate count, n, and active subset sizes; fold fits can dominate. Candidate-pool coverage is an approximation in the decision search. Predictively interchangeable proxies can be acceptable and still count as wrong under exact support recovery. This makes the method scientifically useful but lower priority for the particular individual-BVS contest.

5. SMC WITH BLiP FOR RESOLUTION-ADAPTIVE DISCOVERY

Source: Spector and Janson, Controlled Discovery and Localization of Signals via Bayesian Linear Programming, JASA 120(549), 460-471 (2025 issue; online 2024), DOI 10.1080/01621459.2024.2347667. Entire 12-page published paper read, together with the complete 62-page arXiv v4 including appendices. The arXiv appendices are an earlier version, not a claim to have retrieved the final journal supplement. https://doi.org/10.1080/01621459.2024.2347667 https://arxiv.org/pdf/2203.17208v4

Idea. When the posterior identifies a correlated neighborhood but cannot localize one variable reliably, report the smallest defensible signal-containing group. SMC remains the posterior engine; BLiP is the decision layer. This is particularly well matched to uncertainty spread among substitutes, although it does not solve individual variable identification merely by changing the reporting unit.

Concrete adaptation. For a prespecified, bounded family of groups G, estimate

```
p_G = P(S intersect G is nonempty | y)
    by sum_i W_i 1{S_i intersect G is nonempty}.
```

Do not compute p_G as 1-product_j(1-PIP_j) unless independence is established. Choose mutually disjoint discoveries with a resolution-weighted objective such as sum_G p_G x_G/|G| and the linear BFDR constraint

```
sum_G (1-p_G-alpha)x_G <= 0,
sum_{G containing j} x_G <= 1 for every j,
x_G in {0,1}.
```

BLiP uses an LP relaxation and resolution of fractional decisions. Audit the final integer solution's feasibility and objective gap. Near-integrality is an empirical property in the source, not a universal worst-case guarantee. A feasible budgeted rounding heuristic can retain the input BFDR guarantee while giving up optimality. Report any such gap rather than describing every solution as optimal.

Cost. Limit group size and total candidate count. Use known local structure, multiresolution groups, or budgeted nearest-neighbor construction. For ordered variables, all intervals of bounded length M give O(pM) candidates. For general regression, ordering must be scientifically justified or replaced by another explicit group construction; unrestricted dense correlation clustering can itself cost O(p^2). Computing group probabilities and solving the sparse decision problem must be timed separately from SMC.

Guarantees and limitations. With exact posterior group probabilities, feasibility controls posterior expected GROUP FDP under the specified model. Approximate SMC probabilities do not make the guarantee exact; nor is conditional Bayesian control automatically frequentist control under a fixed misspecified data-generating law. If every candidate probability has absolute error at most epsilon, the selected nonempty collection's true posterior BFDR is at most its computed BFDR+epsilon. This elementary bound illustrates why uniform group-probability checks matter; it does not provide an unearned epsilon certificate for SMC.

Singleton-only BLiP reduces to the familiar ordinary BFDR selection problem. Gains require groups. Overall group BFDR need not control the subset of singleton discoveries by itself; add a separate singleton constraint if making that claim. Never expand each selected group into all its members and count them as individual discoveries. Report group localization and individual BVS on distinct metrics.

Why include it. This is the most developed recent option here for converting a good joint posterior into informative discoveries when fine identification is uncertain. It is a companion to an individual-support selector if individual BVS remains the primary downstream comparison. Earlier local block certificates and resolution ladders make the concept familiar; BLiP supplies a specific joint optimization and error criterion rather than a new posterior model.

SHARED SCALABILITY AND ACCURACY REQUIREMENTS

The current production implementation explicitly rejects p>30 and encodes models with 32-bit integer shifts. Its dense p-by-p Gram matrix and Np coordinate checks per stage are further bottlenecks. Thus the EXISTING CODE is not already a large-p implementation. Replace model encoding with sparse active sets or an arbitrary-length bitset, make caching bounded and collision-safe, and support active-set matrix algebra without a mandatory full Gram matrix. Mutation may need different full-support proposals at large p; accepting a state requires the correct current bridge density and proposal correction. This review does not claim to have developed or validated a new mutation kernel.

No candidate requires global enumeration during production. That is necessary but not sufficient for practical scaling: difficult modes may require more particles, stages, or mutation effort. Benchmark time, memory, likelihood-score evaluations, active sizes, and independent-run disagreement as p grows. Retain the complete model support unless a support restriction is an explicitly defined scientific prior. Decision-only search restrictions are different and must be labeled as such.

For an exactly scored model, w(S)=p(S)m(S), a useful probability query is

```
pi_hat(S) = w(S)/Z_hat,
pi_hat(S)/pi(S)-1 = Z/Z_hat-1.
```

This works even for a model absent from the final particle set. The equality shows precisely why accurate Z is central to the user's aim. It does not prove Z_hat accurate at large p, identify all important models, or allow enumeration of all 2^p probabilities. When posterior mass is very diffuse, small absolute PIP errors alone cannot certify the evidence or every joint probability.

Also distinguish targeting the exact posterior from exact finite-particle inference. Adaptive temperatures do not automatically inherit every fixed- schedule finite-N unbiasedness theorem. No new unbiasedness theorem is claimed.

PROPOSED VALIDATION, NOT EXECUTED

First use existing small-p saved posterior outputs wherever sufficient. Compare exact-posterior and SMC versions of each decision on the SAME datasets. For new priors, obtain independent target-specific enumerations at feasible p, including all normalization constants. Verify joint pattern/group probabilities as well as Z and PIPs. This isolates decision quality from posterior estimation error.

Prespecify tuning rules using training/calibration data or posterior decision criteria, never held-out simulated truth. Evaluate complete tradeoff curves, not a favorable isolated threshold. Record individual exact recovery, TPR, FDP, model size, predictive loss and variability. For BLiP also report discovery resolution and group errors; do not substitute these for individual metrics.

Next run a p ladder beyond 30, with bounded active-set and decision costs made explicit. Independent repeats and stress designs should include mixed-sign signals, strong correlation, weak signals, and prior misspecification. Repeats diagnose instability but cannot prove absence of common bias. Use the separate benchmark thread's fixed datasets and outputs when available; no competitor computations were duplicated here.

For candidate 2, reject a selection improvement if it is purchased with evidence or PIP failure. The primary acceptance gate remains posterior approximation. For candidates 1,3,4,5, preserve the original posterior outputs and separately measure additional decision cost and uncertainty. Owls runs must retain source and

configuration snapshots, seeds, hashes, launch commands, process/scheduler IDs, timestamps, environment, logs, exit statuses, raw outputs and validation.

SOURCE-READING BOUNDARY

The six underlying works cited above were read in the complete versions stated, not inferred from abstracts or training knowledge. Seven reading records are provided because BLiP has both a published main paper and an earlier full arXiv version. Krishna's complete text was accessible, but the PDF endpoint returned an access challenge; archived web-text parts and the primary URL are supplied. Chandra's full author version was accessible, whereas the final journal PDF was not retrieved. These are version/access distinctions, not claims that only abstracts were inspected. Other acquired papers marked unread are not included as evidentiary support for the five recommendations or in the selected bundle.

Project files consulted include Methodology/SMC_Composite_Likelihood_Methodology.tex, the Experiment_2_BVS_Rerun engine and selection/design scripts, its recovery and agreement reports and validation/audit files, Experiment_ 1 saved reports and results manuscript, Further_directions/FD1_xblocks_hybrid saved findings, and SMC_BVS_Manuscript.tex. Full research checkpoints are retained under .codex/smc_bvs_research. No production source or historical result was changed.